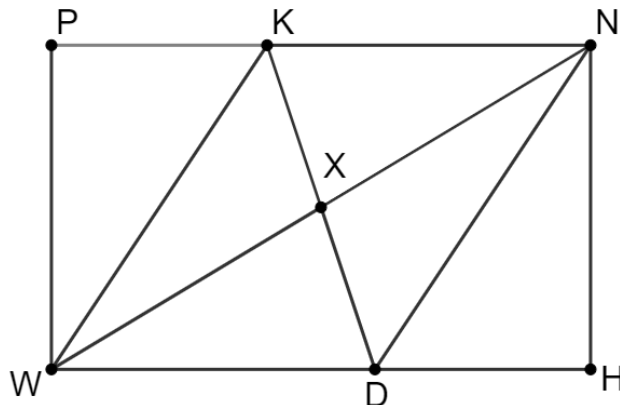


Geometry
Unit 5
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

Quadrilateral $PNHW$ is shown where $\overline{KW} \parallel \overline{ND}$ and X is the midpoint of $\overline{WN}$

Given: $\overline{KW} \parallel \overline{ND}$, X is the midpoint of $\overline{WN}$
Prove: $\overline{WK} \cong \overline{DN}$



Use the given information to answer the questions about the incomplete proof that is shown.

Statement	Reason
1. $\overline{KW} \parallel \overline{ND}$	1. Given
2.	2.
3. X is the midpoint of $\overline{WN}$	3. Given
4.	4.
5.	5.
6.	6.
7. $\overline{WK} \cong \overline{DN}$	7.

1. What is the statement for line 2 of the proof?

$$\angle XKW \cong \angle XDN$$

2. What is the reason for line 2 of the proof?

If two parallel lines are intersected by a transversal, then the alternate interior angles are congruent.

3. What is the statement for line 4 of the proof?

$$\overline{WX} \cong \overline{NX}$$

4. What is the reason for line 4 of the proof?

Definition of midpoint

5. What is the statement for line 5 of the proof?

$$\angle KXW \cong \angle DXN$$

6. What is the reason for line 5 of the proof?

Vertical angles are congruent

7. What is the statement for line 6 of the proof?

$$\triangle XKW \cong \triangle XDN$$

8. What is the reason for line 6 of the proof?

AAS Congruence

9. What is the reason for line 7 of the proof?

Corresponding Parts of Congruent Triangles are Congruent

10. Explain if it is possible to prove that $\triangle WXD \cong \triangle NXK$ using the same given statements.

Yes. Using the proof shown, it can be said that $KX = DX$ by Corresponding Parts of Congruent Triangles are Congruent and $\angle KXN \cong \angle DXW$ because all vertical angles are congruent. Therefore, by SAS Congruence it can be proven that $\triangle WXD \cong \triangle NXK$.